Mohamed Eleraky

AI Engineer & Data Scientist

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience in Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, and Large Language Models (LLMs). Experienced in building end-to-end AI solutions, from data preparation and model development to deployment and real-world applications. Proven track record through internships, leadership roles, competitive achievements, and AI projects spanning computer vision, NLP, multi-agent systems, and predictive analytics. Focused on applying AI technologies to solve complex real-world problems and deliver impactful solutions.

EDUCATION

Bachelor of Computer and Information Science - Ain Shams University 10/2022 – 07/2026
- GPA: 3.5/4.0 Cairo, Egypt

SKILLS

- **Artificial Intelligence (AI):** Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Large Language Models (LLMs), Transformers, Agentic AI, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Prompt Engineering.
- **AI & ML Frameworks:** TensorFlow, PyTorch, Scikit-learn, YOLO, OpenCV, MediaPipe.
- **Data Analysis & Analytics:** Data Cleaning & Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, Statistical Analysis, Data Visualization, Predictive Modeling, SQL, Power BI.
- **Data Science & Deployment Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Streamlit, NiceGUI, Tkinter.
- **Programming Languages:** Python, C++, Java (familiar with C, C#, JavaScript, and TypeScript).
- **Problem Solving:** Data Structures & Algorithms, Competitive Programming.
- **Soft Skills:** Teamwork, Leadership, Communication, Adaptability, Time Management.

EXPERIENCE

NLP Intern - Cellula Technologies 09/2026 – Present
- Working on NLP concepts including text processing, tokenization, RNNs, LSTMs, Transformers, and LLMs. Remote
- Building AI applications using LangChain, LangGraph, agents, and multi-agent systems.
- Applying production tools including Docker, CI/CD, FastAPI, Streamlit, and SQLite.

AI Instructor - Information Technology Institute (ITI) 08/2026 – 09/2026
- Delivered hands-on AI training to junior students, covering Python programming, Data Analysis, and Machine Learning fundamentals. Cairo, Egypt
- Guided students through practical exercises helping them develop AI concepts and problem-solving skills.

Agentic AI Intern – Digital Hub × Orange Partnership 06/2026 – 08/2026
- Completed a 5-week internship focused on Transformers, Prompting & Reasoning, Agentic AI, Multi-Agent Systems, RAG, and Memory Systems. Cairo, Egypt
- Developed practical AI agent solutions in agent deployment, monitoring, and real-world applications.

Computer Vision Intern - Cellula Technologies 02/2026 – 04/2026
- Enhancing and optimizing computer vision models for real-world applications Remote
- Contributing to end-to-end computer vision pipelines to improve model performance, inference efficiency, and model deployment.

AI Internship - CIB Bank 07/2025 – 08/2025
- Analyzed AI applications in the banking sector, focusing on predictive analytics, and financial sustainability. Remote
- Explored AI use cases in financial services and enterprise systems to support informed decision-making.

PROJECTS

iVision: AI-Powered Mobile Assistant for Visually Impaired Users - Graduation Project | FCIS 09/2025 – Present
- Created an offline Android application for navigation, obstacle detection, scene narration, text reading, and voice interaction.
- Implemented adaptive AI pipelines using YOLOv26, MiDaS, and Depth Anything with dynamic model switching based on scene complexity.
- Designed a grid-based obstacle warning system combining object detection and depth estimation for accurate directional alerts.
- Enhanced scene understanding and captioning modules using Places365 and BLIP, alongside OCR-based reading using PaddleOCR.
- Integrated offline navigation via OpenStreetMap and speech interaction through STT/TTS, optimized for real-time mobile deployment.

AI Career Learning Agents - Mansoura Hackathon '26 – [LLMs & NLP] 02/2026 – 02/2026
- Built a multi-agent AI system using LLMs (LangChain) to analyze labor market data and generate personalized learning roadmaps.

- Designed end-to-end pipelines (extraction → analysis → learning plan) with structured outputs and adaptive recommendations.

Brain Tumor Classification – [Deep Learning & Computer Vision] 12/2025 – 02/2026
- Improved an MRI classifier using EfficientNet with fine-tuning to detect 4 tumor types.
- Integrated a GUI using Streamlit with Grad-CAM visualization for tumor region highlighting.

NLP Website Reviews Analysis - FCIS'26 – [Deep Learning & NLP] 11/2025 – 01/2026
- Created a sentiment analysis system for website reviews using Neural Networks from scratch.
- Experimented with multiple architectures (LSTM, GRU, Transformer) to optimize performance.
- Applied NLP preprocessing and trained models on real-world datasets for Kaggle competition.
- Deployed the models using NiceGUI, enabling real-time sentiment prediction and user interaction.

Video Game Sales Prediction - FCIS'25 – [ML] 04/2025 – 06/2025
- Applied preprocessing, feature engineering, and correlation analysis.
- Tested multiple ML models using grid search with (regression, classification) versions.
- Deployed the model using Streamlit for real-time sales predictions.
- Won 3rd place in university competition.

Sign Language Translator - Robotech'25 – [ML & Computer Vision] 04/2025 – 05/2025
- Developed a real-time app for sign language (English/Arabic, letters, numbers, and words).
- Trained ML models and used computer vision to translate camera gestures.
- Integrated ESP and Arduino components to display translated output on an LCD.
- Won 1st and 3rd place in two separate competitions hosted by different universities.

E-Commerce Customer Behavior - DEPI'25 – [Data Analysis & ML] 03/2025 – 05/2025
- Engineered a predictive system using preprocessing, EDA, and feature engineering.
- Tested ML models & optimized via Grid Search & hyperparameter tuning.
- Deployed the model using Streamlit for real-time interaction.

VOLUNTEER

Head of AI – RoboTech Student Activity 10/2025 – 07/2026 Cairo, Egypt
- Leading an AI committee of 100+ members across 3 parallel levels, covering Python, Data Analysis, ML, DL, Computer Vision, and NLP.
- Designed learning tracks, tasks, and workshops with hands-on projects and technical mentoring.
- Supervised 24 AI/ML projects developed in the annual RoboTech Fair event.

Python Judge – GRC Competition | British University in Egypt (BUE) 05/2026 – 05/2026 Cairo, Egypt
- Evaluated and scored participants' Python programming submissions during the competition.
- Assessed solution correctness, efficiency, and code quality according to competition criteria.

AI Roadmap Developer – FIRM Volunteer Initiative 03/2026 – 04/2026 Remote
- Developed a structured AI roadmap covering Python, Data Analysis, ML, DL, Computer Vision, and NLP.
- Curated learning resources, practical workshops, notebooks, and projects for each track.

AI Support & Judge – Robotech Summer Training (Computer Vision) 07/2025 – 08/2025 Cairo, Egypt
- Supported a Computer Vision & IoT summer training program and assisted trainees in practical sessions.
- Participated as a judge in evaluating final student projects, assessing system performance and integration.

COURSES

Agentic AI Track – Digital Egypt Pioneers Initiative Round'5 07/2026 – Present Cairo, Egypt
- Currently undertaking a 6-month program covering Transformers, Generative AI, Prompting & Reasoning, Agentic AI, Multi-Agent Systems, RAG, and Memory Systems.
- Developing AI agents with hands-on training in agent deployment and real-world applications.

Computer Vision Training - National Telecommunication Institute (NTI) 08/2025 – 09/2025 Remote
- Finished a hands-on training program in Computer Vision and Deep Learning.
- Worked on multiple practical tasks and a final project applying CV techniques.

Data Science & AI Track - Digital Egypt Pioneers Initiative Round'2 10/2024 – 06/2025 Cairo, Egypt
- Completed a 6-month program covering Python, SQL, data analysis, visualization, and machine learning.
- Built multiple projects including a final capstone project applying end-to-end data science workflows.

AI & Machine Learning Diploma - AMIT Learning 07/2024 – 02/2025 Cairo, Egypt
- Completed a 7-month diploma covering machine learning, deep learning, NLP, and computer vision.
- Worked on multiple hands-on projects using real-world datasets and AI pipelines.

ACHIEVEMENTS & COMPETITIONS

2nd Place – AZEX 2026 Software Track 05/2026

3rd Place – GRC 2025 Computer Vision & Robotics 05/2025

1st Place – AZEX 2025 Computer Vision & Embedded Systems 05/2025

3rd Place – FCIS 2025 Machine Learning Contest 06/2025

LANGUAGES

Arabic — Native | **English** — Fluent